

Satya Sai Satyavarapu

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Professional Summary

Frontend-focused Software Engineer with 4.5+ years of experience building scalable, high-performance web and mobile applications using React.js, TypeScript, Next.js, and React Native. Experienced in frontend architecture, performance optimization, reusable component systems, and responsive UI development. Additionally experienced in Node.js backend services, real-time systems, and AI-driven integrations for full-stack product development.

Technical Skills

- **Languages:** JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3
- **Frontend Development:** React.js, Next.js, React Native, Redux, Context API, Tailwind CSS, Material UI, List Virtualization
- **Backend Engineering:** Node.js, Express.js, REST APIs, FastAPI, Socket.io, Redis, BullMQ, Queue Management, Webhooks
- **Databases & Storage:** MongoDB (Mongoose ODM), PostgreSQL, Azure Blob Storage
- **AI Integrations:** Google Gemini API, HeyGen API, Multi-modal Media Processing
- **Cloud, DevOps & Tools:** Docker, Azure, Nginx, Vercel, Render, Git, GitHub, CI/CD Concepts
- **Testing & Architecture:** Jest, React Testing Library, Postman, Event-driven Architecture, Rate Limiting, Authentication, Cron Jobs

Key Achievements

- Reduced JS bundle size by **50%** improving Lighthouse score from **55** → **85**.
- Engineered **40+ reusable UI components** accelerating feature delivery by **30%**.
- Built AI processing pipeline handling **100+ jobs/day**.
- Improved real-time UX by replacing polling with **WebSocket streaming**.
- Resolved production **memory leaks** improving application uptime.

Professional Experience

Caprus IT Private Limited

Software Engineer

Hyderabad, India

May 2024 – Present

- Expanded from frontend development to backend ownership by building APIs, MongoDB schemas, and queue-based workflows for product features.
- Engineered a **50% reduction in initial JavaScript payload size** (from 1.8 MB to under 900 KB) using route-based code splitting and dependency pruning, **elevating Lighthouse Performance scores from the mid-50s to the mid-80s**.
- Built an event-driven workflow using BullMQ and Redis to manage AI processing with retry handling.
- Established real-time job tracking using **WebSockets and Socket.io**, replacing polling and improving live status updates for users.
- Resolved **WebSocket memory leaks** by improving connection handling, increasing application stability in production.

Caprus IT Private Limited

Frontend Developer

Hyderabad, India

Jan 2022 – Apr 2024

- Shipped high-performance, responsive cross-platform user interfaces across web (React) and mobile (React Native) ecosystems, maintaining strict feature parity and design consistency.
- Developed a reusable component library with **40+ shared UI components**, reducing duplicate code and improving development speed.
- Improved application loading and rendering performance on lower-end Android devices using **optimized Redux selectors, memoization, lazy loading, and list virtualization**.
- Worked closely with QA, product, and backend teams to finalize API requirements and speed up feature integration.

Product Development & Projects

Aauti Marketplace — MERN Stack, Redis, BullMQ, WebSockets, HeyGen API

Company Project

- Architected an AI video generation workflow handling **50–100+ daily jobs** using Redis, BullMQ, and background workers.
- Created webhook handling with retry and fallback logic to handle third-party API failures reliably.
- Built backend APIs to securely connect with external AI services while protecting API keys.

Aauti School (K-12 Platform) — React, React Native, Redux, RESTful APIs

Company Project

- Developed dynamic curriculum flows and state management for grade-based learning experiences.
- Integrated curriculum APIs to support dynamic lesson planning and content updates.

- Built a multi-modal content moderation platform analyzing text, PDFs, images, and videos using Gemini API.
- Developed configurable risk scoring and flagged-content workflows to detect unsafe content.
- Deployed Dockerized setup ensuring consistent environments across development and production.

Education

Kakinada Institute of Engineering & Technology <i>MBA, Information Technology</i>	Kakinada, India <i>Sept 2019 – Sep 2021</i>
Aditya Degree College <i>Bachelor of Science, Computer and Information Sciences</i>	Kakinada, India <i>July 2015 – Apr 2018</i>